

RW4 Promo-State and Discount Models

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

incidence_rows=1

type_rows=1

depth_rows=1

asymmetry_rows=1322

Table: Promo_State_Incidence

standardized_type	model_scope	model_unreliable_flag	reason
ALL	unavailable	binomial_attempt_failed	1 Promo-incidence binomial ...

Table: Promo_State_Type

standardized_type	model_scope	model_unreliable_flag	reason
ALL	unavailable	multinomial_attempt_failed	1 Promo-state multinomial m...

Table: Promo_State_Depth

standardized_type	model_scope	model_unreliable_flag	reason
ALL	unavailable	depth_attempt_failed	1 Promo-depth conditional m...

Table: Asymmetry_Observed_vs_Baseline

segment_name	standardized_type	farm_gate_source	reconstruction_variant	chain_direction	stage_from	stage_to	model_family	observed_sr_coef	baseline_sr_coef	delta_sr_coef	observed_lr_coef	baseline_lr_coef	delta_lr_coef	observed_ect_coef	baseline_ect_coef	delta_ect_coef	same_sign_lr	same_sign_ect	pseudo_asymmetry_likely
Вершки	cream	initial	linear	forward	ProZorro	Retail	NARDL	0.1855195387021934	0.1007392494158206	-0.0847802892863728	0.2943590807532827	0.146246553131392	-0.1481125276218907	-0.7295217442383444	-0.7945546025645894	-0.06503285832624506	True	True	0
	cream	initial	linear	forward	ProZorro	Retail	NARDL	0.1855195387021934	0.1007392494158206	-0.0847802892863728	0.2943590807532827	0.146246553131392	-0.1481125276218907	-0.7295217442383444	-0.7945546025645894	-0.06503285832624506	True	True	0
	cream	initial	linear	forward	FarmGateUA	ProZorro	ARDL	27.40581164655441	27.40581164655441	0.0	-1.563367219701413	-1.563367219701413	0.0	nan	nan	nan	True	False	0
	cream	initial	linear	forward	FarmGateUA	ProZorro	ARDL	27.40581164655441	27.40581164655441	0.0	-1.563367219701413	-1.563367219701413	0.0	nan	nan	nan	True	False	0
	cream	initial	linear	forward	FarmGateUA	ProZorro	NARDL	-39.44395541420697	-39.44395541420697	0.0	-18.22659908428248	-18.22659908428248	0.0	-1.394427676720398	-1.394427676720398	0.0	True	True	0
	cream	initial	linear	forward	FarmGateUA	ProZorro	NARDL	-39.44395541420697	-39.44395541420697	0.0	-18.22659908428248	-18.22659908428248	0.0	-1.394427676720398	-1.394427676720398	0.0	True	True	0
	cream	initial	linear	forward	FarmGateUA	Retail	NARDL	-8.772023103981054	-4.075323960016448	4.696699143964606	-1.852043589957451	-2.798583866922371	-0.94654027696492	-0.7869901901005358	-0.9482306245503481	-0.1612404344498123	True	True	1
	cream	initial	linear	forward	FarmGateUA	Retail	NARDL	-8.772023103981054	-5.709143818937397	3.062879285043657	-1.852043589957451	-4.32428951476429	-2.472245924806839	-0.7869901901005358	-0.9225873084949123	-0.1355971183943765	True	True	1
	cream	initial	linear	forward	FarmGateUA	Retail	NARDL	-8.772023103981054	-5.709143818937397	3.062879285043657	-1.852043589957451	-4.32428951476429	-2.472245924806839	-0.7869901901005358	-0.9225873084949123	-0.1355971183943765	True	True	1
	cream	initial	linear	forward	FarmGateUA	Retail	NARDL	-8.772023103981054	-4.075323960016448	4.696699143964606	-1.852043589957451	-2.798583866922371	-0.94654027696492	-0.7869901901005358	-0.9482306245503481	-0.1612404344498123	True	True	1
	cream	initial	linear	reverse	Retail	ProZorro	NARDL	1.562917141304475	2.310060540554998	0.7471433992505232	-0.2420696049936481	0.694874012870825	0.936943617864473	-1.11930546143876	-1.131840757655833	-0.012535296217072833	False	True	1
	cream	initial	linear	reverse	Retail	ProZorro	NARDL	1.562917141304475	2.310060540554998	0.7471433992505232	-0.2420696049936481	0.694874012870825	0.936943617864473	-1.11930546143876	-1.131840757655833	-0.012535296217072833	False	True	1
	cream	initial	linear	reverse	ProZorro	FarmGateUA	NARDL	0.001294370750072191	0.001294370750072191	0.0	0.02102571537973896	0.02102571537973896	0.0	-0.1349163065634996	-0.1349163065634996	0.0	True	True	0
	cream	initial	linear	reverse	ProZorro	FarmGateUA	NARDL	0.001294370750072191	0.001294370750072191	0.0	0.02102571537973896	0.02102571537973896	0.0	-0.1349163065634996	-0.1349163065634996	0.0	True	True	0
	cream	initial	linear	reverse	Retail	FarmGateUA	NARDL	0.006744790717442998	-0.01054007989158433	-0.01728487060902733	0.0979596312980813	0.1881569946873936	0.09019736338931231	-0.144872313268366	-0.1099961288446253	0.0348761844237407	True	True	0
	cream	initial	linear	reverse	Retail	FarmGateUA	NARDL	0.006744790717442998	-0.0113459641210052	-0.0180907548384482	0.0979596312980813	0.1011228731519556	0.003163241853874302	-0.144872313268366	-0.08797236879732441	0.05689994447104159	True	True	0
	cream	initial	linear	reverse	Retail	FarmGateUA	NARDL	0.006744790717442998	-0.0113459641210052	-0.0180907548384482	0.0979596312980813	0.1011228731519556	0.003163241853874302	-0.144872313268366	-0.08797236879732441	0.05689994447104159	True	True	0
	cream	initial	linear	reverse	Retail	FarmGateUA	NARDL	0.006744790717442998	-0.01054007989158433	-0.01728487060902733	0.0979596312980813	0.1881569946873936	0.09019736338931231	-0.144872313268366	-0.1099961288446253	0.0348761844237407	True	True	0
Молоко	питне	initial	linear	forward	ProducerUA	ProZorro	NARDL	-39.35627563072685	-39.35627563072685	0.0	-1.428284989733576	-1.428284989733576	0.0	-0.9847510668431191	-0.9847510668431191	0.0	True	True	0
Молоко	питне	milk	initial	forward	ProZorro	Retail	NARDL	-0.05150967266593062	-0.07759645603786203	-0.026086783371931405	-0.04955189374108957	-0.1875189372472206	-0.13796704350613104	-0.5842457751880188	-0.5984997219947372	-0.014253946806718432	True	True	0
Молоко	питне	milk	initial	forward	ProZorro	Retail	NARDL	-0.05150967266593062	-0.07759645603786203	-0.026086783371931405	-0.04955189374108957	-0.1875189372472206	-0.13796704350613104	-0.5842457751880188	-0.5984997219947372	-0.014253946806718432	True	True	0
Молоко	питне	milk	initial	forward	ProducerUA	Retail	NARDL	-15.29852457315202	16.08250489949674	31.381029472648763	1.143785547528498	5.19149651385324	4.0477109663247415	-0.7215120205461834	-0.9142138640633395	-0.19270184351715614	True	True	1
Молоко	питне	milk	initial	reverse	Retail	ProZorro	NARDL	-3.377883072190306	-4.3193244960139	-0.9414414238235942	-0.8065514462952509	-1.65425420802465	-0.8477027617293992	-0.9583680273094293	-1.048219149760883	-0.08985112245145377	True	True	1
Молоко	питне	milk	initial	reverse	Retail	ProZorro	NARDL	-3.377883072190306	-4.3193244960139	-0.9414414238235942	-0.8065514462952509	-1.65425420802465	-0.8477027617293992	-0.9583680273094293	-1.048219149760883	-0.08985112245145377	True	True	1
Молоко	питне	milk	initial	reverse	ProZorro	ProducerUA	ARDL	-0.003981249234879858	-0.003981249234879858	0.0	0.08556001807837454	0.08556001807837454	0.0	nan	nan	nan	True	False	0
Молоко	питне	milk	initial	reverse	ProZorro	ProducerUA	NARDL	-0.06869734434387867	-0.06869734434387867	0.0	-0.7347687591369463	-0.7347687591369463	0.0	-0.1809708253238888	-0.1809708253238888	0.0	True	True	0
Молоко	питне	milk	initial	reverse	Retail	ProducerUA	NARDL	-0.2289575426204663	0.03043734523875096	0.25939488785921727	-4.567563531007199	4.632234498859097	9.199798029866296	-0.05896759892425604	-0.04125585800661566	0.017711740917640384	False	True	1
Сир	твердый	hard_cheese	initial	forward	FarmGateUA	ProducerUA	NARDL	0.9718885163453532	0.9718885163453532	0.0	4.164046497719572	4.164046497719572	0.0	-0.02099956516295825	-0.02099956516295825	0.0	True	True	0
Сир	твердый	hard_cheese	initial	forward	ProducerUA	ProZorro	ARDL	2.533791668492441	2.533791668492441	0.0	0.8515901678330565	0.8515901678330565	0.0	nan	nan	nan	True	False	0
Сир	твердый	hard_cheese	initial	forward	ProducerUA	ProZorro	NARDL	-81.10378249455037	-81.10378249455037	0.0	-3.429536449371221	-3.429536449371221	0.0	-0.913365423696856	-0.913365423696856	0.0	True	True	0
Сир	твердый	hard_cheese	initial	forward	ProZorro	Retail	ARDL	-0.008082951427838936	-0.2316252020709719	-0.22354225064313296	-0.1583981708763298	0.08260811044262882	-0.07579006043370097	nan	nan	nan	True	False	1
Сир	твердый	hard_cheese	initial	forward	ProZorro	Retail	ARDL	-0.008082951427838936	-0.2316252020709719	-0.22354225064313296	-0.1583981708763298	0.08260811044262882	-0.07579006043370097	nan	nan	nan	True	False	1
Сир	твердый	hard_cheese	initial	forward	ProZorro	Retail	NARDL	0.01249324517355312	-0.3638721758900579	-0.376365421063611	0.2033362629705365	-0.4370611614962356	-0.6403974244667721	-1.086095722865538	-0.9455579020345951	0.1405378208309428	False	True	1
Сир	твердый	hard_cheese	initial	forward	ProZorro	Retail	NARDL	0.01249324517355312	-0.3638721758900579	-0.376365421063611	0.2033362629705365	-0.4370611614962356	-0.6403974244667721	-1.086095722865538	-0.9455579020345951	0.1405378208309428	False	True	1
Сир	твердый	hard_cheese	initial	forward	FarmGateUA	ProZorro	NARDL	-68.77473824351345	-68.77473824351345	0.0	-11.37819556032246	-11.37819556032246	0.0	-1.331056770131037	-1.331056770131037	0.0	True	True	0
Сир	твердый	hard_cheese	initial	forward	FarmGateUA	ProZorro	NARDL	-68.77473824351345	-68.77473824351345	0.0	-11.37819556032246	-11.37819556032246	0.0	-1.331056770131037	-1.331056770131037	0.0	True	True	0
Сир	твердый	hard_cheese	initial	forward	ProducerUA	Retail	NARDL	17.92245839749829	51.00085056474553	33.07839216724724	5.795116967333372	8.708316733703773	2.9131997663704015	-0.9697285021614469	-1.115149434966051	-0.145420932804604	True	True	1
Сир	твердый	hard_cheese	initial	forward	FarmGateUA	Retail	ARDL	-16.12941410918768	14.40821465960636	30.537628768794043	-3.273094954784757	-1.000896839545215	2.272198115239542	nan	nan	nan	True	False	1
Сир	твердый	hard_cheese	initial	forward	FarmGateUA	Retail	ARDL	-16.12941410918768	17.8105304655541	33.93994457474178	-3.273094954784757	-0.4393484941635429	2.833746460621214	nan	nan	nan	True	False	1
Сир	твердый	hard_cheese	initial	forward	FarmGateUA	Retail	ARDL	-16.12941410918768	17.8105304655541	33.93994457474178	-3.273094954784757	-0.4393484941635429	2.833746460621214	nan	nan	nan	True	False	1

Table: Discount_Strategy_Synthesis

question	answer	evidence
Do promotions behave like...	yes	Baseline Silpo ECM-style ...
Does pseudo-asymmetry wea...	yes	Observed vs baseline coef...
Do effective prices adjus...	mixed	share_observed_ect_faster...
Does discount depth react...	mixed	Promo_State_Depth stores ...
Do retail shocks transmit...	yes	reverse_core_rows=119
Is the retail-to-farmgate...	yes	reverse_robust_share=1.00
Is transmission stronger ...	yes	product_core_rows=301; av...
Do specific retailer bran...	yes	brand_core_rows=55

Observed vs Baseline Short-Run Delta

